

Storage tank

Storage tanks are containers that hold liquids, compressed gases (gas tank; or in U.S.A "pressure vessel", which is not typically labeled or regulated as a storage tank) or mediums used for the short- or long-term storage of heat or cold.^[1] The term can be used for reservoirs (artificial lakes and ponds), and for manufactured containers. The usage of the word tank for reservoirs is uncommon in American English but is moderately common in British English. In other countries, the term tends to refer only to artificial containers.



Spherical gas tank farm in the petroleum refinery in Karlsruhe MiRO

In the USA, storage tanks operate under no (or very little) pressure, distinguishing them from *pressure vessels*. Storage tanks are often cylindrical in shape, perpendicular to the ground with flat bottoms, and a fixed flangible or floating roof. There are usually many environmental regulations applied to the design and operation of storage tanks, often depending on the nature of the fluid contained within. Above-ground storage tanks (ASTs) differ from underground storage tanks (USTs) in the kinds of regulations that are applied. Above ground storage tanks can be used to hold materials such as petroleum, waste matter, water, chemicals, and other hazardous materials, all while meeting strict industry standards and regulations.^[2]

Reservoirs can be covered, in which case they may be called covered or underground storage tanks or reservoirs. Covered water tanks are common in urban areas.

Storage tanks are available in many shapes: vertical and horizontal cylindrical; open top and closed top; flat bottom, cone bottom, slope bottom and dish bottom. Large tanks tend to be vertical cylindrical, or to have rounded corners transition from vertical side wall to bottom profile, to easier withstand hydraulic hydrostatically induced pressure of contained liquid. Most *container tanks* for handling liquids during transportation are designed to handle varying degrees of pressure.

A large storage tank is sometimes mounted on a lorry (truck) or on an articulated lorry trailer, which is then called a tanker.

In the U.S., air emissions, which correspond to losses for usable liquid commodities and/or salable products, are typically required to undergo air quality permitting under the Federal Clean Air Act. Quantification of potential emissions for permitting purposes is most often accomplished by applying emission equations published in Chapter 7.1 of the U.S. EPA's AP-42: Compilation of Air Emission Factors.

Contents

Special features

For refineries

Other types of tank

- Atmospheric

- High pressure

- Thermal storage tanks